

IP LEET 2019

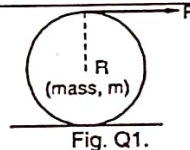
QUESTION BOOKLET SERIES : A

Max. Marks : 600 No. of Questions : 150 Time : 150 min

Instructions :

- Write your roll No. on the Question Booklet and also on the OMR Answer Sheet in the space provided and no where else.
- Enter the Code No. of Question Booklet on the OMR answer Sheet. Darken the corresponding bubbles with Black Ball Point/Black Gel Pen.
- Do not make any identification mark on the Answer Sheet or Question Booklets.
- To open the Question Booklet remove the seal gently when asked to do so.
- Please check that this Question Booklet contains 150 questions. In case of any discrepancy, inform the Assistant Superintendent within 10 minutes of the start of test.
- Each question has four alternative answer (A, B, C, D) of which only one is correct. For each question, darken only one bubble (A or B or C or D), whichever you think is the correct answer, on the Answer Sheet with Black Point/Black Gel Pen. There shall be negative marking for wrong answers.
- If you do not want to answer a question, leave all the bubbles corresponding to that question blank in the Answer Sheet. No marks will be deducted in such cases.
- Darken the bubbles in the OMR Answer Sheet according to the Serial No. of the questions given in the Question Booklet.
- If you want to change an already marked answer, erase the shade in the darkened bubble completely.
- For rough work only the blank sheet at the end of the Question Booklet be used.
- The Answer Sheet is designed for computer evaluation. Therefore, if you do not follow the instructions given on the Answer Sheet, it may make evaluation by the computer difficult. Any resultant loss to the candidate on the above account, i.e. not following the instructions completely, shall be of the candidate only.
- After the test, hand over the Question Booklet and the Answer Sheet to the Assistant Superintendent on duty.
- In no case the Answer Sheet, the Question Booklet, or its part or any material copied/noted from this Booklet is to be taken out of the examination hall. Any candidate found doing so would be expelled from the examination.
- A candidate who creates disturbance of any kind or changes his/her seat or is found in possession of any paper possibly of any assistance or found given or receiving assistance or found using any other unfair means during the examination will be expelled from the examination by the Centre Superintendent/Observer whose decision shall be final.
- Telecommunication Equipment such as mobile phones, pager, cellular phone, wireless set, scanner, camera or any electronic gadget etc., is not permitted inside the examination hall. Use of calculators is not allowed.
- The candidates will not be allowed to leave the Examination Hall/Room before the expiry of the allotted time.

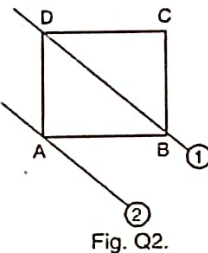
1. A tangential force F acts at the top of a thin spherical shell of mass m and radius R . Find the acceleration of the shell if it rolls without slipping.



- (A) $a = \frac{5F}{6m}$ (B) $a = \frac{6F}{5m}$
(C) $a = \frac{3F}{2m}$ (D) $a = \frac{3F}{4m}$

2. Let I_1 and I_2 be the moment of inertia of a uniform square plate about axes as shown in Fig. Q2. Then, the ratio $I_1 : I_2$ is

- (A) $1 : \frac{1}{7}$ (B) $1 : \frac{12}{7}$
(C) $1 : \frac{7}{12}$ (D) $1 : 7$



3. If the radius of earth contracts to half of its present value without being change in its mass. Then, new duration of the day will be

- (A) 6 h (B) 5 h
(C) 9 h (D) No change

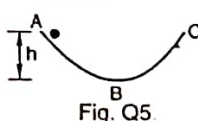
4. A S.H.M. is represented by the equation,

$$x = 10 \sin \left(\pi t + \frac{\pi}{6} \right).$$

What will be its maximum velocity in m/s?

- (A) 5π (B) 6π
(C) 10π (D) None of these

5. A solid ball rolls down a parabolic curve path ABC from a height h as shown in Fig. below. Portion AB of the path is rough while BC is smooth.



How high will the ball climb in BC ?

- (A) $\frac{5}{7}h$ (B) $\frac{5}{6}h$
(C) $\frac{7}{5}h$ (D) $35h$

6. If the resultant of two forces P and Q acting at an angle (α) with P , then

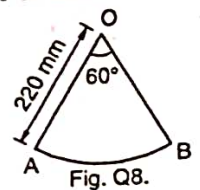
- (A) $\tan \alpha = \frac{P \sin \theta}{P + Q \cos \theta}$ (B) $\tan \alpha = \frac{P \cos \theta}{P + Q \cos \theta}$
(C) $\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$ (D) $\tan \alpha = \frac{Q \cos \theta}{P + Q \cos \theta}$

7. A smooth cylinder lying on a horizontal plane is an example of which type of equilibrium?

- (A) Stable equilibrium (B) Unstable equilibrium
(C) Neutral equilibrium (D) None of these

8. The centre of gravity of the lamina as shown in Fig. Q8 of radius 220 mm, is given as

- (A) 130 mm (B) 140 mm
(C) 120 mm (D) 100 mm



9. Theorem of perpendicular axis is used in obtaining the MOI of a

- (A) Triangular lamina (B) Square lamina
(C) Circular lamina (D) Semicircular lamina

10. Which of the following is TRUE about the maximum efficiency of screw jack?

- (A) It depends upon the weight lifted or effort applied.
(B) It depends upon the angle of friction.
(C) It depends upon both weight lifted and angle of friction.
(D) None of these.

11. A certain weight lifting machine of velocity ratio 30 cm can lift a load of 1500 N with the help of 125 N effort. The machine is in nature.

- (A) Reversible (B) Irreversible
(C) Both (A) and (B) (D) None of these

12. The maximum mechanical advantage of a lifting machine is given as

- (A) $1/m$ (B) m (C) m^2 (D) $\sqrt{m}$

13. A framed structure is perfect, if the number of members are $(2j - 3)$, where j is the number of joints.

- (A) equal to (B) less than
(C) greater than (D) either (B) or (C)

14. The concurrent for equilibrium of coplanar non-concurrent force system are

- (A) $\Sigma F_x = 0, \Sigma F_y = 0$ (B) $\Sigma F_x = 0, \Sigma M = 0$
(C) $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M = 0$ (D) All of the above

15. A particle starts from rest. The ratio of distances covered by it in the 3rd and 5th seconds of its motion is

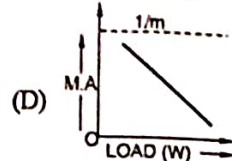
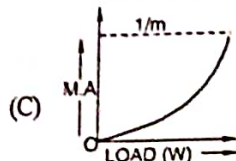
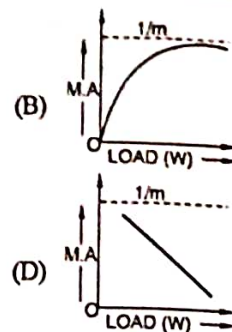
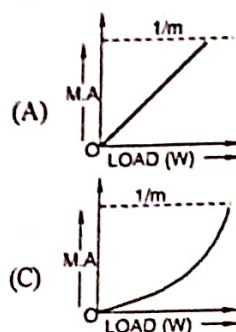
- (A) 9 : 5 (B) 5 : 9 (C) 4 : 9 (D) 9 : 4

16. A particle starting from rest, moves of in a straight line, with equation of motion equal to $s = t^3 - 2t^2 + 3$. The acceleration of particle after 3 seconds (in m/s^2) is

- (A) 14 (B) 15 (C) 26 (D) 19

17. A projectile is fired upwards at an angle of 30° with a velocity of 40 m/s. The time taken to reach the ground by projectile, after instant of firing is

- (A) 4.08 sec (B) 8.04 sec
(C) 80 sec (D) 9.8 sec
18. A bullet is fired with a velocity 100 m/s at an angle of 45° , with the horizontal. How high the bullet will reach?
(A) 255 m (B) 250 m (C) 240 m (D) 200 m
19. A weight of 1 kN is lifted by an effort of 125 N by second system of pulleys, having 5 pulleys in each block. The amount of effort wasted in friction is (N)
(A) 20 (B) 240 (C) 25 (D) 10
20. In a compound lever, the leverages of all the simple levers is
(A) Added (B) Subtracted
(C) Multiplied (D) Divided
21. The dimensional formula for force is
(A) MT^{-2} (B) MLT^{-2} (C) ML^{-1} (D) MLT
22. All the forces in the space diagram are named using the
(A) D'Alembert principle (B) Varignon's theorem
(C) Bow's notation (D) None of these
23. The coefficient of restitution for inelastic bodies is
(A) zero (B) between zero and one
(C) one (D) infinite
24. The loss of kinetic energy due to direct impact of two bodies depends upon :
(A) Mass of two bodies.
(B) Initial velocities of two bodies.
(C) Final velocities of two bodies.
(D) Both (A) and (B).
25. A pile of negligible mass is driven by a hammer of mass 200 kg. If the pile is driven 500 mm into the ground, when the hammer falls from a height of 4 m. The average force of resistance of ground (N)
(A) 17640 (B) 17000 (C) 17650 (D) 17630
26. The mass MOI of a uniform thin rod about one of the ends perpendicular to the length.
(A) $\frac{Ml^2}{12}$ (B) $\frac{4Ml^2}{3}$ (C) $\frac{Ml^2}{8}$ (D) $\frac{Ml^2}{6}$
27. The unit of mass moment of inertia is
(A) m^4 (B) mm^4 (C) N-m (D) $kg \cdot m^2$
28. The centre of gravity of hemisphere lies at a distance of from its base measured along the vertical radius.
(A) $\frac{3r}{8}$ (B) $\frac{3}{8r}$ (C) $\frac{8r}{3}$ (D) $\frac{8}{3r}$
29. Which one of the following is TRUE in the context of Analysis of frame?
(A) The ends of members are fixed.
(B) The ends of members are pin jointed.
(C) Self weight of members are neglected.
(D) Both options (B) and (C)
30. The angle between the resultant reaction and the normal to the plane on which the motion of the body is impending is called as :
(A) Angle of repose (B) Angle of limiting friction
(C) Angle of obliquity (D) All of these
31. Which one of the following graph is TRUE for the variation of mechanical advantage with respect to load.



32. The velocity ratio for inclined plane is given as
(A) $\csc \theta$ (B) $\tan \theta$ (C) $\sin \theta$ (D) $\sec \theta$
33. If two objects of 30 kg and 10 kg move with equal kinetic energy, then what is ratio of magnitudes for linear momentum?
(A) $\sqrt{3}:1$ (B) $1:\sqrt{3}$ (C) $1:3\sqrt{3}$ (D) $1:3$
34. What is the average resistance required to stop a truck of mass 600 kg in a distance of 30 m, if initial speed is 30 m/sec?
(A) 8 kN (B) 9 kN (C) 9.5 kN (D) 9 N
35. A ball dropped from a wall of height h travels a distance of 50 m in last two seconds before landing. What is the height of wall from which the ball was dropped?
(A) 120.15 m (B) 127.37 m
(C) 183.48 m (D) None of above
36. What is the distance of centroid with respect to diagonal shown in diagram below?
(A) $a/\sqrt{3}$
(B) $a/\sqrt{2}$
(C) $a/\sqrt{18}$
(D) $3a/\sqrt{2}$

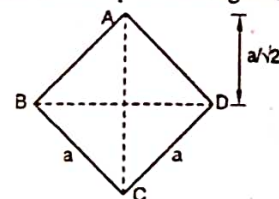


Fig. Q36.

37. What is the relation between tight side and slack side of a flat belt?
(A) (Tight side tension $\times$ Slack side tension) = $e^{\mu \theta}$
(B) (Tight side/Slack side tension) = $e^{\mu \theta}$
(C) (Tight side tension + Slack side tension) = $e^{\mu \theta}$
(D) None of these
38. Which formula is used to calculate the value of angle of static friction (ϕ_s)?
(A) $\tan^{-1}(\mu_s)$ (B) $\sin^{-1}(\mu_s)$
(C) $\cos^{-1}(\mu_s)$ (D) None of the above
39. Three resistances are joined together to form a letter Y, as shown in Fig. Q. 39. If the potentials of the terminals A, B and C are 6V, 3V and 2V respectively, then the potential of the point O will be
(A) 4V (B) 3V
(C) 2.5V (D) 0V

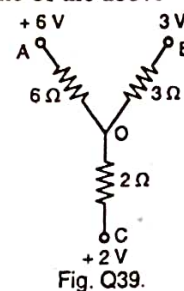
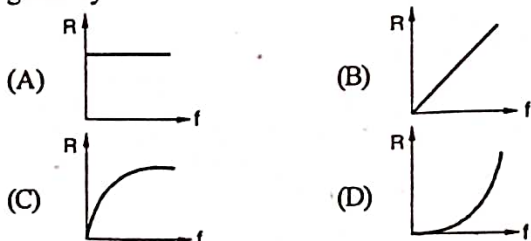


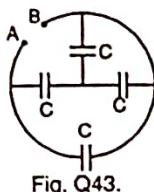
Fig. Q39.

40. A resistor R has a power of dissipation P with cell voltage E . The resistor is cut in n equal parts and all the parts are connected in parallel with same cell. The new power dissipation is
(A) nP (B) nP^2 (C) n^2P (D) n/P

41. The electric field in a region is given by $\vec{E} = a\hat{i} + b\hat{j}$ where a and b are constants. The net flux passing through square area of side ' l ' parallel to $y-z$ plane is
(A) a^2l (B) al^2 (C) a^2l^2 (D) al
42. A proton, a deuteron and an alpha particle are accelerated through potentials of V , $2V$ and $4V$ respectively. Their velocities will bear a ratio of
(A) $1:1:1$ (B) $1:\sqrt{2}:1$
(C) $\sqrt{2}:1:1$ (D) $1:1:\sqrt{2}$
43. Find the equivalent capacitance between points A and B as shown in Fig. Q43.
(A) $\frac{4C}{3}$ (B) $\frac{3C}{2}$
(C) $4C$ (D) $\frac{5C}{3}$
44. What will be the smallest value of magnetic field B that can be setup at the equator to permit a proton of speed 10^7 m/s to circulate around the earth? (Assume $R = 6.4 \times 10^6$ m, $m_p = 1.67 \times 10^{-27}$ kg)
(A) 1.6×10^{-8} T (B) 1.5×10^{-8} T
(C) 1.2×10^{-8} T (D) 1.9×10^{-8} T
45. Large transformers, when used for some times, becomes very hot and are cooled by circulating oil. The heating of the transformer is due to
(A) The heating effect of electric current alone.
(B) Hysteresis loss alone.
(C) Both heating effect of current and hysteresis loss.
(D) Intense sunlight at noon.
46. Mirage is due to
(A) Unequal heating of different parts of the atmosphere.
(B) Magnetic disturbance areas in atmosphere.
(C) Depletion of ozone layer in atmosphere.
(D) Equal heating of different parts of the atmosphere.
47. Rectifiers are used to convert.....
(A) Direct current to alternating current.
(B) Alternating current to direct current.
(C) High voltage to low voltage.
(D) Low voltage to high voltage.
48. Rainbow formation is due to
(A) Absorption of sunlight in minute water droplets.
(B) Diffusion of sunlight through water droplets.
(C) Ionization of water deposits.
(D) Refraction and reflection of sunlight by water droplets.
49. The term $\cos \phi$ in AC circuit is called
(A) Form factor (B) Phase factor
(C) Power factor (D) Quality factor
50. The correct variation of resistance R with frequency F is given by



51. In Fig. Q.51, if the current i decreases at a rate α , then $V_A - V_B$ is
(A) zero (B) $-\alpha L$
(C) αL (D) No relation exist
52. Colligative properties are proportional to
(A) Congo Rubin Number (B) Van't Hoff factor
(C) Boyle's law (D) None of these
53. The temperature at which the molarity of pure water is equal to its molality is :
(A) 273 K (B) 298 K
(C) 277 K (D) None of these
54. Isopiestic solutions have
(A) Same vapor pressure (B) Same osmotic pressure
(C) Same freezing point (D) Same boiling point
55. Solubility of deliquescent substance in water is generally.
(A) High (B) Low
(C) Moderate (D) Can't be predicated
56. If the kinetic energy of an electron is increased 4 times, the wavelength of the de-broglie wave associated with it would become :
(A) 4 times (B) 2 times (C) $1/2$ times (D) $1/4$ times
57. The ratio of radii of first orbits of H, He^+ , and Li^{2+} is :
(A) $1:2:3$ (B) $6:3:2$ (C) $1:4:9$ (D) $9:4:1$
58. Born Haber cycle is generally used
(A) To calculate lattice energy by indirect experiment
(B) Calorific value
(C) Number of moles of solution
(D) None of these
59. For the reaction, $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$, the rate of reaction is
(A) $\frac{1}{2} \frac{d}{dt} [\text{N}_2\text{O}_5]$ (B) $2 \frac{d}{dt} [\text{N}_2\text{O}_5]$
(C) $\frac{1}{4} \frac{d}{dt} [\text{NO}_2]$ (D) $4 \frac{d}{dt} [\text{NO}_2]$
60. For the hypothetical reactions, the equilibrium constant (k) values are given
 $\text{A} \rightleftharpoons \text{B}; k_1 = 2$ $\text{B} \rightleftharpoons \text{C}; k_2 = 4$,
 $\text{C} \rightleftharpoons \text{D}; k_3 = 3$
The equilibrium constant (k) for the reaction $\text{A} \rightleftharpoons \text{D}$ is
(A) 48 (B) 6 (C) 2.7 (D) 24
61. The frequency of radiation having wave number 10 m^{-1} is:
(A) 10 s^{-1} (B) $3 \times 10^7 \text{ s}^{-1}$
(C) $3 \times 10^{11} \text{ s}^{-1}$ (D) $3 \times 10^9 \text{ s}^{-1}$
62. Ostwald dilution law holds good for
(A) Weak electrolytes (B) Strong electrolytes
(C) Both weak and strong electrolytes
(D) None of these
63. Which of the following is/are amphoteric ?
(A) $\text{NH}_3, \text{H}_2\text{O}, \text{H}_2\text{S}, \text{CO}_2$ (B) $\text{NH}_3, \text{H}_2\text{O}, \text{CH}_3\text{COOH}$
(C) $\text{H}_2\text{O}, \text{H}_2\text{S}, \text{HCl}, \text{NaOH}$ (D) None of these
64. Which of the following is a lewis acid ?
(A) BF_3 (B) Na^+
(C) SeF_4 (D) All of the above
65. Oxidation state of carbon in HCOOH will be
(A) +1 (B) +2 (C) -4 (D) 0



66. In button cells, the electrolyte is a paste of

- (A) $\text{Zn}(\text{OH})_2 + \text{KOH}$ (B) $\text{ZnO} + \text{MnO}$
(C) $\text{Zn}(\text{OH})_2 + \text{MnO}$ (D) $\text{ZnO} + \text{KOH}$

67. Which enzyme converts cane sugar into glucose and fructose ?

- (A) Distase (B) Invertase
(C) Zymase (D) Maltase

Directions (68 -71) : Find the related word/letter/number to complete the analogy.

68. House : Rent :: Capital : ?

- (A) Interest (B) Investment
(C) Country (D) Money

69. NUMBER : UNBMRE :: GHOST : ?

- (A) HOGST (B) HOGTS
(C) HGSOT (D) HGOST

70. 18 : 30 :: 36 : ?

- (A) 64 (B) 66 (C) 54 (D) 62

71. From the given alternatives, select the set which is most like the given set. Given set (4, 10, 15)

- (A) (3, 6, 12) (B) (2, 8, 10)
(C) (5, 12, 18) (D) (7, 10, 18)

Directions (72-76) : Find the ODD man out.

72. (A) SKY - stars (B) Moon - Planets
(C) Stadium - Players (D) University - Students

73. (A) BFJQ (B) RUZG
(C) GJOV (D) ILQX

74. (A) Plant - Flower (B) Chair - Sofa
(C) Face - Eye (D) Tree - Stem

75. (A) 12-96 (B) 13-117
(C) 15-120 (D) 16-128

76. (A) AJKL (B) IBCD
(C) ORQP (D) UFGH

Directions (77-78) Which one of the given response would be a meaning full order of the following.

77. 1. Fruit 2. Flower
3. Seed 4. Pollination

5. Bud
(A) 1, 2, 3, 4, 5 (B) 4, 2, 5, 3, 1
(C) 5, 2, 4, 3, 1 (D) 5, 2, 4, 1, 3

78. 1. Infant 2. Old
3. Adult 4. Adolescent

5. Child
(A) 5, 4, 3, 2, 1 (B) 3, 4, 2, 1, 5
(C) 1, 5, 4, 3, 2 (D) 2, 3, 4, 5, 1

79. In a certain language, BUTTER is coded as CVUUF, BREAD is coded as CSFBE, then how COFFEE is coded?

- (A) DPGGFF (B) GGDPFF
(C) GDPGFF (D) FFDPGG

80. In a coding system PEN is written as NZO and BARK as CTSL. How can we write PRANK in that coding system?

- (A) NZTOL (B) CSTZN
(C) NSTOL (D) NTSLO

81. E is the sister of B, A is the father of C and B is the son of C. How is A related to E?

- (A) Grand father (B) Grand daughter
(C) Father (D) Great-grandfather

82. If '+' stands for multiplication, 'x' stands for division, '-' stands for addition and ÷ stands for subtraction, what

would the following equation stands for ?
 $20 - 8 \times 4 \div 3 + 2 = ?$

- (A) 41 (B) 19 (C) 16 (D) 18

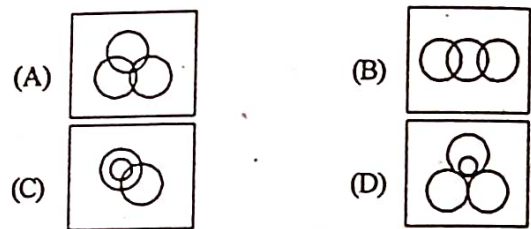
83. Mohan starts from point A and walks 1 km towards south, turns left and walks 1 km. Then, he turns left again and walks 1 km. Now he is facing towards.

- (A) East (B) West
(C) North (D) South-west

84. How many days were there from January 2, 1995 to March 15, 1995?

- (A) 73 (B) 74 (C) 71 (D) 35

85. Which diagram represents the relationship among females, mothers and doctors ?



86. Which number is common in the square, ellipse and triangle?

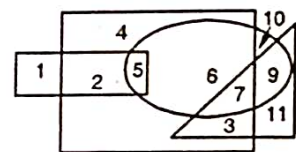


Fig. Q86.

- (A) 1 (B) 5 (C) 6 (D) 7

87. A is older than B but younger than C. D is younger than E but older than A. If C is younger than D. Who is the oldest of all?

- (A) A (B) C
(C) D (D) E

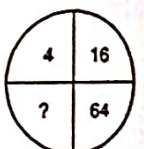


Fig. Q87.

88. Select the missing number from the given responses.

- (A) 16 (B) 52 (C) 112 (D) 256

89. BC, GH, LM ?

- (A) PQ (B) RS (C) QR (D) OP

90. In a certain code language, 'BAD' is written as '7' and 'SAP' is written as '9'. How is 'BAN' written in that code language ?

- (A) 8 (B) 3 (C) 4 (D) 6

91. Three positions of a cube are shown below. What will come opposite to face containing I ?

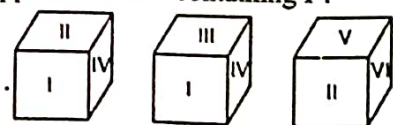


Fig. Q 91

- (A) VI (B) IV (C) II (D) V

92. Which of the following fraction is the smallest?

$$\frac{7}{6}, \frac{7}{9}, \frac{4}{5}, \frac{5}{7}$$

- (A) 7/6 (B) 7/9 (C) 4/5 (D) 5/7

93. The HCF of two numbers is 15 and their LCM is 300. If one of the number is 60, the other is

- (A) 50 (B) 75 (C) 65 (D) 100

94. The value of $1 - \frac{a}{1 - \frac{1}{1 + \frac{a}{1 - a}}}$ is

- (A) a (B) $1 - a$ (C) 1 (D) 0
95. $0.01024^{1/5}$ is equal to
(A) 4.0 (B) 0.04 (C) 0.4 (D) 0.00004
96. The average weight of five persons sitting in a boat is 38 kg. The average weight of the boat and the persons sitting in the boat is 52 kg. What is the weight of boat?
(A) 228 kg (B) 122 kg
(C) 232 kg (D) 242 kg
97. If $a : b = 7 : 9$ and $b : c = 15 : 7$, then what is $a : c$?
(A) 5 : 3 (B) 3 : 5
(C) 7 : 21 (D) 7 : 15
98. The sum of squares of two positive numbers is greater than their product by 28. If the ratio of the numbers is 2 : 3, find the numbers.
(A) 4 and 6 (B) 6 and 9
(C) 8 and 12 (D) None of these
99. If 50% of $(x - y) = 30\%$ of $(x + y)$ then, what percent of x is y ?
(A) 25% (B) $33\frac{1}{3}\%$ (C) 40% (D) 400%
100. By selling 33 meters of cloth, a person gains the cost of 11 meters. Find his gain %.
(A) $33\frac{1}{3}\%$ (B) $33\frac{1}{2}\%$ (C) 33% (D) $34\frac{1}{3}\%$
101. If sweets are bought at 15 for a rupee. How many must be sold for a rupee to gain 25%?
(A) 10 (B) 11 (C) 12 (D) 8
102. What single discount is equivalent to two successive discounts of 20% and 15%?
(A) 35% (B) 32% (C) 34% (D) 30%
103. A sum of ₹1600 gives a simple interest of ₹202 in 2 years and 3 months. The rate of interest per annum is :
(A) 5.5% (B) 8% (C) 7% (D) 6%
104. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of first train is :
(A) 70 km/hr (B) 75 km/hr
(C) 84 km/hr (D) 87.5 km/hr
105. The angle of elevation of a ladder leaning against a wall is 60° and the foot of the ladder is 4.6 m away from the wall. The length of ladder is
(A) 2.3 m (B) 4.6 m (C) 7.8 m (D) 9.2 m
106. If $\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a + b)$, then
(A) $a + b = 1$ (B) $a - b = 1$
(C) $a = b$ (D) $a^2 - b^2 = 1$
107. If A and B are invertible matrices of order 3, $|A| = 2$ and $|(AB)^{-1}| = -\frac{1}{6}$, then value of $|B|$ is
(A) -3 (B) 2 (C) 1 (D) -1

108. The derivative of $\sin^2(x^2)$ w.r.t x^2 is
(A) $\sin(2x^2)$ (B) $\tan(x^2)$
(C) $\cos(x^2)$ (D) None of these
109. The integral of $\int \frac{e^x(x-3)}{(x-1)^3} dx$ is
(A) $\frac{e^x}{(x-1)^3} + c$ (B) $\frac{e^x}{(x-1)^4} + c$
(C) $\frac{(x-3)^2 e^x}{(x-1)^3} + c$ (D) $\frac{e^x}{(x-1)^2} + c$
110. If $4 \sin^{-1} x + \cos^{-1} x = \pi$, then the value of x .
(A) $\frac{1}{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) 0
111. If $e^y(x+1) = 1$, then the value of $\frac{dy}{dx} =$
(A) e^y (B) e^{-y} (C) $-e^y$ (D) $-e^{-y}$
112. The mean of the numbers obtained on throwing a die having written 1 on three faces, 2 on two faces and 5 on one face is
(A) 1 (B) 2 (C) 5 (D) $\frac{8}{3}$
113. If the area of triangle is 35 sq. units with vertices $(2, -6)$, $(5, 4)$ and $(k, 4)$. Then k is
(A) 12 (B) -2 (C) -12, -2 (D) 12, -2
114. The slope of the normal to the curve $y = 2x^2 + 3 \sin x$ at $x = 0$ is
(A) 3 (B) $\frac{1}{3}$ (C) -3 (D) $-\frac{1}{3}$
115. The function $f(x) = 3x + 17$ is
(A) Strictly increasing on R
(B) Strictly decreasing on R
(C) Neither increasing or decreasing on R
(D) None of the above
116. The number of non-zero integral solutions of the equation $|1 - i|^x = 2^x$ is
(A) 2 (B) 1 (C) 3 (D) 0
117. If $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$ then the value of x is
(A) 64 (B) 65 (C) 32 (D) 16
118. The value of $\tan 11^\circ \tan 17^\circ \tan 79^\circ \tan 73^\circ$ is
(A) $\frac{1}{2}$ (B) 0 (C) 1 (D) $\frac{1}{\sqrt{2}}$
119. The maximum value of $\sin^4 \theta + \cos^4 \theta$ is
(A) $\frac{1}{3}$ (B) 1 (C) 2 (D) 3
120. The area of region bounded by the curve $y^2 = x$ and the lines $x = 1$, $x = 4$ and the x -axis is
(A) $\frac{15}{3}$ (B) 6 (C) 1 (D) $\frac{14}{3}$
121. The integral of $\frac{\cos x}{\sqrt{1 + \sin x}}$ is
(A) $\sqrt{1 + \sin^2 x}$ (B) $2\sqrt{1 + \sin x}$
(C) $\sqrt{1 + 2 \sin x}$ (D) None of these
122. The projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$ is
(A) 0 (B) 1 (C) 3 (D) 2
123. The value of $\lim_{x \rightarrow 0} x \sec x$ is
(A) 1 (B) 0 (C) 2 (D) $\frac{1}{2}$

124. The value of $\lim_{x \rightarrow 0} \frac{\sin ax}{bx}$
 (A) b/a (B) ab (C) a/b (D) 1
125. The distance between parallel lines $15x + 8y - 34 = 0$, and $15x + 8y + 31 = 0$ is
 (A) $\frac{65}{17}$ (B) $\frac{65}{18}$ (C) $\frac{65}{15}$ (D) $\frac{15}{16}$
126. In a group of students, 100 students know Hindi, 50 know English and 25 know both. Each of the students knows either Hindi or English. How many students are there in a group?
 (A) 120 (B) 125 (C) 135 (D) 150
127. Which of the following differential equation has $y = x$ as one of its particular solution?
 (A) $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} + xy = x$
 (B) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + xy = x$
 (C) $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} + xy = 0$
 (D) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + xy = 0$
128. A line has direction ratios $-18, 12, -4$, then its direction cosines are
 (A) $\frac{-9}{11}, \frac{-2}{11}, \frac{6}{11}$ (B) $\frac{-9}{2}, \frac{6}{9}, \frac{9}{11}$
 (C) $0, 0, 1$ (D) $\frac{-9}{11}, \frac{6}{11}$ and $\frac{-2}{11}$
129. The length of latus rectum for $x^2 = -16y$ is
 (A) 4 (B) 8 (C) -4 (D) 16
130. If $P(A) = \frac{1}{2}$, $P(B) = 0$, then $P(A/B)$ is equal to.
 (A) 0 (B) $1/2$ (C) Not defined (D) 1
131. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to
 (A) 6 (B) ± 6 (C) -6 (D) 0
132. If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, then $\frac{d^2y}{dx^2} =$
 (A) $\frac{\sec^2 t}{at}$ (B) $\frac{\sec t}{at}$
 (C) $\frac{\cos t}{at}$ (D) None of these
133. The value of integral $\int_0^1 \tan^{-1} \left(\frac{2x-1}{1+x-x^2} \right) dx$ is
 (A) 1 (B) 0
 (C) -1 (D) $\frac{\pi}{4}$
134. The general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is
 (A) $e^x + e^{-y} = c$ (B) $e^x + e^y = c$
 (C) $e^{-x} + e^y = c$ (D) None of these
135. The unit vector in the direction of the vector $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ is
 (A) $\frac{1}{\sqrt{6}}\hat{i} + \frac{1}{\sqrt{6}}\hat{j} + \frac{2}{\sqrt{6}}\hat{k}$ (B) $2\hat{i} + 3\hat{j} + 5\hat{k}$
 (C) $\hat{i} + \hat{j} + \hat{k}$ (D) None of these
136. The direction cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$
 (A) $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{5}{\sqrt{14}}$ (B) $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$
 (C) $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{5}{\sqrt{14}}$ (D) All of the above.
137. $\sin(\tan^{-1} x)$, $|x| < 1$ is equal to
 (A) $\frac{x}{\sqrt{1-x^2}}$ (B) $\frac{1}{\sqrt{1-x^2}}$
 (C) $\frac{1}{\sqrt{1+x^2}}$ (D) $\frac{x}{\sqrt{1+x^2}}$
138. $\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$ is equal to
 (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$
 (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
139. If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ then transpose of square matrix A is
 (A) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \cos \theta & \sin \theta \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 (C) $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
140. If A and B are square matrix then $(AB)' =$
 (A) $A'B'$ (B) $B'A'$
 (C) AB' (D) $A'B$
141. USB stands for
 (A) Universal Serial Bus (B) Unity Serial Bus
 (C) Universal Security Bus (D) Unity Security Bus
142. Spread sheets are saved with extension
 (A) .xls (B) .doc
 (C) .pdf (D) .mp3
143. Which one of the following option of Microsoft word 2007 has been replaced by File Tab in word 2010?
 (A) Save As (B) Open
 (C) New (D) Backup
144. Which one is cyber crime?
 (A) Sending Email (B) Online purchasing
 (C) Online gambling (D) Chatting with friends.
145. Which of the following is NOT type of internet working?
 (A) Intranet (B) Extranet
 (C) Subnet (D) Internet
146. HTML is used for
 (A) Website (B) System software
 (C) Application software (D) Utilities software.

147. Which of the following is NOT a system software ?
 (A) Operating system (B) Compiler
 (C) Assembler. (D) Pagemaker
148. The first developed virus for computer was
 (A) Norton (B) PC-clic
 (C) Brain virus (D) Boot-P
149. The first web browser developed was
 (A) LYNX (B) MOSAIC
 (C) Mozilla fire fox (D) Google chrome
150. Data communication system spanning across the world is
 (A) LAN (B) WAN
 (C) MAN (D) None of the above

ANSWER

- | | | | | |
|----------|----------|----------|----------|----------|
| 1. (B) | 2. (C) | 3. (A) | 4. (C) | 5. (A) |
| 6. (C) | 7. (C) | 8. (B) | 9. (B) | 10. (D) |
| 11. (B) | 12. (A) | 13. (A) | 14. (C) | 15. (B) |
| 16. (A) | 17. (A) | 18. (A) | 19. (D) | 20. (C) |
| 21. (B) | 22. (C) | 23. (A) | 24. (D) | 25. (A) |
| 26. (B) | 27. (D) | 28. (A) | 29. (D) | 30. (B) |
| 31. (B) | 32. (A) | 33. (A) | 34. (B) | 35. (B) |
| 36. (C) | 37. (B) | 38. (A) | 39. (B) | 40. (C) |
| 41. (B) | 42. (D) | 43. (D) | 44. (A) | 45. (C) |
| 46. (A) | 47. (B) | 48. (D) | 49. (C) | 50. (A) |
| 51. (B) | 52. (B) | 53. (C) | 54. (A) | 55. (A) |
| 56. (C) | 57. (B) | 58. (A) | 59. (C) | 60. (D) |
| 61. (D) | 62. (A) | 63. (B) | 64. (D) | 65. (B) |
| 66. (D) | 67. (B) | 68. (A) | 69. (C) | 70. (B) |
| 71. (C) | 72. (B) | 73. (A) | 74. (B) | 75. (B) |
| 76. (C) | 77. (D) | 78. (C) | 79. (A) | 80. (C) |
| 81. (A) | 82. (C) | 83. (C) | 84. (A) | 85. (C) |
| 86. (D) | 87. (D) | 88. (D) | 89. (C) | 90. (A) |
| 91. (A) | 92. (D) | 93. (B) | 94. (D) | 95. (C) |
| 96. (B) | 97. (A) | 98. (A) | 99. (A) | 100. (A) |
| 101. (C) | 102. (B) | 103. (C) | 104. (D) | 105. (D) |
| 106. (A) | 107. (A) | 108. (A) | 109. (D) | 110. (C) |
| 111. (B) | 112. (B) | 113. (D) | 114. (D) | 115. (A) |
| 116. (D) | 117. (A) | 118. (C) | 119. (B) | 120. (D) |
| 121. (B) | 122. (A) | 123. (B) | 124. (C) | 125. (A) |
| 126. (B) | 127. (C) | 128. (D) | 129. (D) | 130. (C) |
| 131. (B) | 132. (A) | 133. (B) | 134. (A) | 135. (A) |
| 136. (B) | 137. (D) | 138. (B) | 139. (C) | 140. (B) |
| 141. (A) | 142. (A) | 143. (D) | 144. (C) | 145. (C) |
| 146. (A) | 147. (D) | 148. (C) | 149. (B) | 150. (B) |

HINTS AND SOLUTIONS FOR THE SELECTED QUESTIONS :

Sol. 1. (B) Let f be the friction force between the spherical shell and horizontal level.

For translational motion, $F + f = ma$... (1)

For rotational motion, $F \times R - f \times R$

$$= I\alpha = \frac{I \times a}{R} \quad \left[\because \alpha = \frac{a}{R} \right]$$

$$F - f = \frac{I \times a}{R^2} \quad \dots (2)$$

Adding Eqn. (1) and (2), we get

$$2F = \left(m + \frac{I}{R^2} \right) \times a = \left(m + \frac{2m}{3} \right) a = \frac{5}{3} m \times a$$

$$F = \frac{5}{3 \times 2} \times ma = \frac{5}{6} ma$$

$$a = \frac{6F}{5m} \quad \text{Ans.}$$

or

Sol. 2. (C)

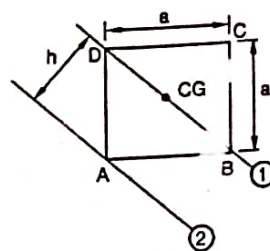


Fig. A2.

a = Side of square plate,
 h = Distance between axes (1) and (2)

We know that, MOI of square plate about C.G is given by

$$I_1 = \frac{a^4}{12} \quad \dots (1)$$

$$h = \frac{1}{2} \times \text{length of diagonal } AC$$

$$= \frac{1}{2} \times \sqrt{2} \times a = \frac{a}{\sqrt{2}}$$

Applying parallel axis theorem,

$$I_2 = I_{CG} + A \times h^2$$

$$I_2 = I_1 + a^2 \times \left(\frac{a}{\sqrt{2}} \right)^2$$

$$I_2 = \frac{a^4}{12} + \frac{a^4}{2} = \frac{7a^4}{12} \quad \dots (2)$$

Ratio of MOI $I_1 : I_2 = 1 : \frac{7}{12}$ [Using (1) and (2)]

Sol. 3. (A) Present Angular momentum of earth,

$$L_1 = I\omega = \frac{2}{5} MR^2 \times \omega$$

As the radius changes, angular momentum will also change, so, new angular momentum due to change in radius of earth.

$$L_2 = \frac{2}{5} M \left(\frac{R}{2} \right)^2 \times \omega = \frac{2}{5} \times M \times \frac{R^4}{4} \times \omega'$$

As the external torque is zero, angular momentum must be conserved,

$$L_1 = L_2$$

$$\frac{2}{5} MR^2 \times \omega = \frac{2}{5} \times \frac{MR^2}{4} \times \omega'$$

$\Rightarrow$

$$\omega' = 4\omega$$

Or Time period, $T' = \frac{1}{4} T$ [$\because \omega \propto \frac{1}{T}$]

$$T' = \frac{1}{4} \times 24 = 6h$$

$$T' = 6h \quad \text{Ans.}$$

Sol. 4. (C) Comparing with standard equation,

$$x = A \sin(\omega t + \phi)$$

we get,

$$A = 10m$$

$$\omega = \pi s^{-1}$$

$$\phi = \pi/6$$

Maximum velocity, $V_{\max} = \omega \times A = 10\pi \text{ m/s}$ (Ans.)

Sol. 5. (A) At B, total kinetic energy = mgh

Ratio of rotational to translation kinetic energy,

$$\frac{k_R}{k_T} = \frac{2}{5} \left[\because k_R = \frac{2}{7} mgh, k_T = \frac{5}{7} mgh \right]$$